1: Uniformly sample an index set $I \subseteq [m]$ and strings $Y_1, \cdots Y_m \in \{0,1\}^{N/m}$.

2: For all $j \in [m]$: Let $Z_j = \begin{cases} Y_j & \text{if } j \in I \\ Y_j \oplus X & \text{otherwise} \end{cases}$

3: Run algorithm $\mathcal{B}$ on $(Z_1, \cdots, Z_m)$, get the outcome $a = (a_1, \cdots, a_m)$.

4: **if** $\exists t \in \{0,1\}, \forall j \in [m] : a_j = \begin{cases} \mathsf{XOR}(Y_j) & \text{if } j \in I \\ \mathsf{XOR}(Y_j) \oplus t & \text{otherwise} \end{cases}$ **then**

5: Output $1 \oplus t$.

6: **else**

7: Output a uniformly random bit.

8: **end if**